Retrieval-Augmented Generation for Fact-Heavy QA Case Study

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Comparing Parametric-only, Vanilla RAG, and Advanced RAG

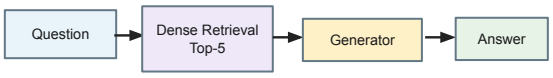
1. MOTIVATION

Small language models struggle with fact-heavy multi-hop QA because parametric memory is limited and hard to update. Retrieval can ground generation in external evidence, but only if the retrieved context is relevant and complete. We study when RAG improves factual consistency and what latency, memory, and robustness trade-offs it introduces.

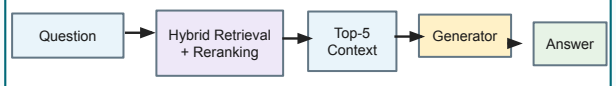
2. SYSTEMS

1. Parametric-only

2. Vanilla RAG



3. Advanced RAG



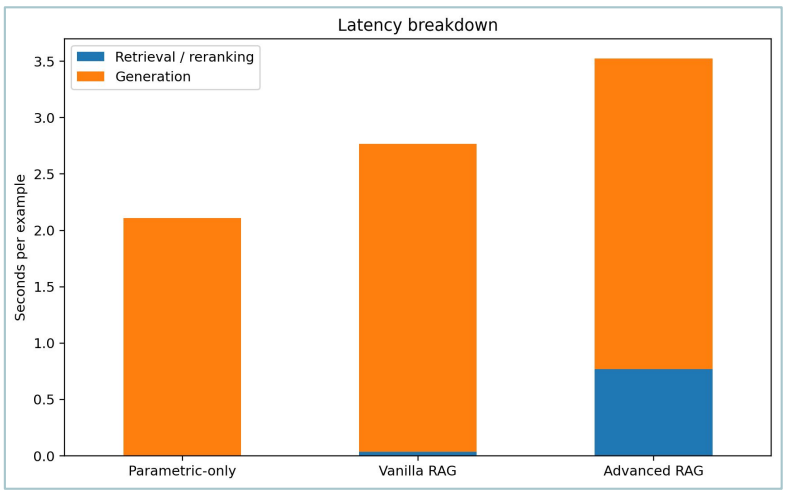
3. EXPERIMENTAL SETUP

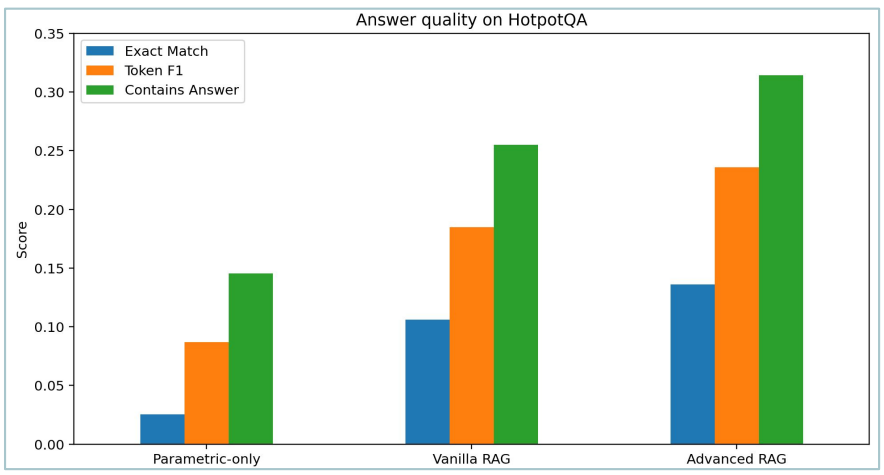
Dataset: HotpotQA
Evaluation set: 2,000 examples
Gate calibration set: 500 examples
Generator: Qwen2.5-0.5B-Instruct
Generator size: 1,884.6 MB
Advanced retriever models:
- -sentence-transformers/all-MiniLM-L6-v2
- -cross-encoder/ms-marco-MiniLM-L-6-v2

Indexed documents: 49,774
Device: MPS
Metrics: EM, Token F1, Contains Answer, Answer Loss, Median PPL, Latency, Hit@5, SupportRecall@5

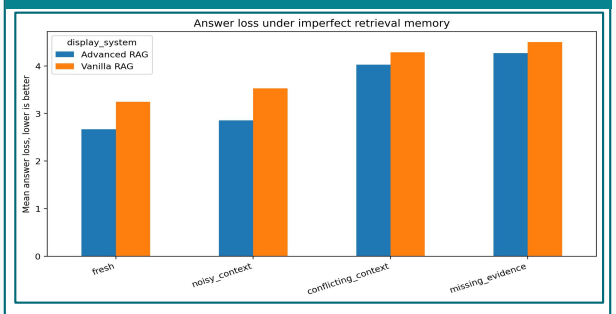
4. MAIN RESULTS

System	Exact Match	Token F1	Contains Answer	Mean Answer Loss	Median PPL	Mean Total Latency (s)
Parametric-only	0.0255	0.0869	0.1455	3.9455	29.65	2.111
Vanilla RAG	0.1060	0.1849	0.2550	3.2459	12.40	2.769
Advanced RAG	0.1360	0.2359	0.3145	2.6662	5.36	3.524





5. ROBUSTNESS: Loss & F1



Condition	Vanilla RAG	Advanced RAG
fresh	0.1849	0.2359
noisy_context	0.1646	0.2098
conflicting_context	0.1140	0.1321
missing_evidence	0.0989	0.1153

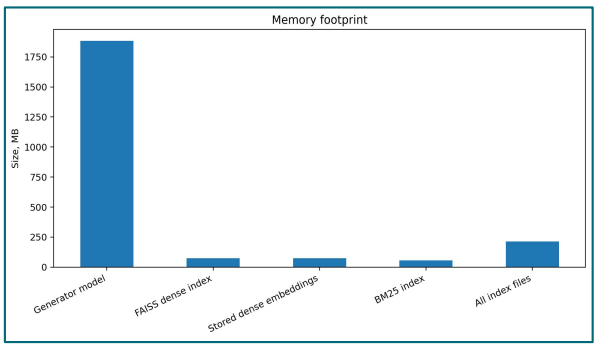
Advanced RAG remains better than vanilla RAG across all four retrieval conditions, but both systems degrade under conflicting or missing evidence.

6. NAIVE QUALITY GATE ABLATION

System	Exact Match	Token F1	Contains Answer
Advanced RAG with naive gate	0.0595	0.1118	0.1730
Advanced RAG without active gate	0.1360	0.2359	0.3145

Threshold used: 5.7721
Quality-gated rate: 0.517
The naive gate rejected many examples and hurt answer quality.

7. MEMORY FOOTPRINT



Component	Size (MB)
Generator model	1,884.6
FAISS dense index	72.9
Stored dense embeddings	72.9
BM25 index	56.2
All index files	212.5

8. KEY FINDINGS / CONCLUSIONS

Retrieval improves factual QA over the parametric-only baseline.

Advanced RAG achieves the best answer quality: Exact Match 0.1360 and Token F1 0.2359.

Retrieved evidence lowers mean answer loss from 3.9455 to 2.6662 and lowers median PPL from 29.65 to 5.36.

Better quality comes with higher latency: 2.111 s for parametric-only, 2.769 s for vanilla RAG, and 3.524 s for advanced RAG.

Robustness drops when retrieval memory is noisy, conflicting, or missing evidence.

A naive rerank-score gate is not useful here because it hurts answer quality.